

Continuity: ϵ - δ and Sequential Characterisations

Math Foundations — Node 11

First, an example

Before the abstract definition, here is the smallest non-trivial calculation: $f(x) = 2x + 1$ is continuous at $x = 3$.

Note that $f(3) = 7$. Fix $\epsilon > 0$ and choose $\delta = \epsilon/2$. Then whenever $|x - 3| < \delta$,

$$|f(x) - 7| = |(2x + 1) - 7| = |2x - 6| = 2|x - 3| < 2\delta = \epsilon.$$

Read aloud: the absolute value of $f(x)$ minus seven equals the absolute value of two x minus six, which equals two times the absolute value of x minus three, which is less than two delta, which equals epsilon.

So $|x - 3| < \delta \Rightarrow |f(x) - 7| < \epsilon$, and f is continuous at 3. The rest of this lesson generalises this template.

1 Definitions

Let (X, d_X) and (Y, d_Y) be metric spaces. We routinely specialise to $X = Y = \mathbb{R}$ with $d(x, y) = |x - y|$.

Definition 1 (Continuity at a point). *A function $f : X \rightarrow Y$ is continuous at $c \in X$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that*

$$d_X(x, c) < \delta \implies d_Y(f(x), f(c)) < \epsilon.$$

Read aloud: whenever the distance from x to c is less than delta, the distance from $f(x)$ to $f(c)$ is less than epsilon.

f is continuous on X if it is continuous at every point.

Definition 2 (Uniform continuity). *f is uniformly continuous on X if for every $\epsilon > 0$ there exists $\delta > 0$ such that for all $x, y \in X$,*

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \epsilon.$$

Read aloud: for every pair x, y , whenever their distance is less than delta, the distance between $f(x)$ and $f(y)$ is less than epsilon.

The single δ must serve every pair of points; the quantifier order distinguishes this from pointwise continuity.

Definition 3 (Lipschitz continuity). *f is Lipschitz with constant $L \geq 0$ if*

$$d_Y(f(x), f(y)) \leq L d_X(x, y) \quad \text{for all } x, y \in X.$$

Read aloud: the distance from $f(x)$ to $f(y)$ is at most L times the distance from x to y , for all x and y in X .

The trivial chain of implications holds:

$$\text{Lipschitz} \implies \text{uniformly continuous} \implies \text{continuous}.$$

Read aloud: Lipschitz implies uniformly continuous, which implies continuous.

None of the converses hold: $\sqrt{x}$ on $[0, 1]$ is uniformly continuous but not Lipschitz; x^2 on $\mathbb{R}$ is continuous but not uniformly continuous.

2 Main theorem

Theorem 4 (ϵ - δ and sequential continuity are equivalent). *Let $f : X \rightarrow Y$ be a function between metric spaces and let $c \in X$. The following are equivalent.*

(A) *f is continuous at c in the sense of Definition 1.*

(B) *For every sequence $(x_n) \subseteq X$ with $x_n \rightarrow c$, we have $f(x_n) \rightarrow f(c)$ in Y .*

Proof. **(A) $\Rightarrow$ (B).** Assume f is ϵ - δ continuous at c and let (x_n) be any sequence with $x_n \rightarrow c$. Fix $\epsilon > 0$. By (A), choose $\delta > 0$ such that

$$d_X(x, c) < \delta \implies d_Y(f(x), f(c)) < \epsilon. \quad (\star)$$

Read aloud: whenever the distance from x to c is less than delta, the distance from $f(x)$ to $f(c)$ is less than epsilon; call this implication star.

Since $x_n \rightarrow c$, there exists $N \in \mathbb{N}$ with $d_X(x_n, c) < \delta$ for all $n \geq N$. For such n , $(\star)$ gives $d_Y(f(x_n), f(c)) < \epsilon$. Hence $f(x_n) \rightarrow f(c)$, establishing (B).

(B) $\Rightarrow$ (A). We argue by contrapositive: suppose f is *not* ϵ - δ continuous at c . Then there exists $\epsilon_0 > 0$ such that for every $\delta > 0$, some $x \in X$ satisfies $d_X(x, c) < \delta$ yet $d_Y(f(x), f(c)) \geq \epsilon_0$.

Specialise to $\delta = 1/n$ for each $n \in \mathbb{N}$: pick $x_n \in X$ with

$$d_X(x_n, c) < \frac{1}{n} \quad \text{and} \quad d_Y(f(x_n), f(c)) \geq \epsilon_0.$$

Read aloud: the distance from x_n to c is less than one over n , and the distance from $f(x_n)$ to $f(c)$ is at least epsilon-naught.

The first inequality forces $x_n \rightarrow c$. The second forces $f(x_n) \not\rightarrow f(c)$ (otherwise the distances would eventually be smaller than ϵ_0). This violates (B), establishing the contrapositive and hence (B) $\Rightarrow$ (A). $\square$

3 Consequences

Lemma 5 (Composition). *If $f : X \rightarrow Y$ is continuous at c and $g : Y \rightarrow Z$ is continuous at $f(c)$, then $g \circ f$ is continuous at c .*

Proof. By Theorem 4, continuity is preserved along sequences: if $x_n \rightarrow c$ then $f(x_n) \rightarrow f(c)$, hence $g(f(x_n)) \rightarrow g(f(c))$. Apply Theorem 4 in the reverse direction. $\square$

Theorem 6 (Heine–Cantor, stated). *If $K \subseteq \mathbb{R}^d$ is compact and $f : K \rightarrow \mathbb{R}$ is continuous, then f is uniformly continuous on K .*

The proof (omitted) follows the contrapositive recipe of Theorem 4: failure of uniformity yields sequences $(x_n), (y_n)$ with $|x_n - y_n| \rightarrow 0$ yet $|f(x_n) - f(y_n)| \geq \epsilon_0$; compactness extracts a convergent subsequence, contradicting pointwise continuity.

4 Example

Example 7 (x^2 at $x = 2$). Fix $\epsilon > 0$. For $|x - 2| < 1$ we have $|x + 2| < 5$, so

$$|x^2 - 4| = |x - 2| |x + 2| < 5 |x - 2|.$$

Read aloud: *the absolute value of x-squared minus four equals the absolute value of x minus two times the absolute value of x plus two, which is less than five times the absolute value of x minus two.*

Choose $\delta = \min(1, \epsilon/5)$. Then $|x - 2| < \delta \Rightarrow |x^2 - 4| < \epsilon$, verifying Definition 1.